A few directions in optimal transport

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Matchings

Example

- distribution of buyers,
- distribution of sellers,
- how to match the buyers and sellers

Example

- pixels in one photo
- pixels in another photo
- how to match these pixels in different photos. (questions in image processing)

Example

- a distribution of cells in different cell types in an organism today
- a distribution of cells in different cell types in an organism tomorrow
- how the organism change from today to tomorrow.

Optimal transport as optimal matching

- Source and target domains X, Y, respectively.
- μ, ν source and target measures of *equal mass*, on X and Y, respectively.
  - μ is a distribution of $x \in X$, ν is a distribution of $y \in Y$.

A **transportation plan** π is a matching between μ and ν.
It is a measure on $X \times Y$, which satisfies

$$\int_Y d\pi(\cdot, y) = \mu(\cdot) \quad \text{(the first marginal of } \pi \text{ is } \mu),$$

$$\int_X d\pi(x, \cdot) = \nu(\cdot) \quad \text{(the second marginal of } \pi \text{ is } \nu),$$

$$\pi \geq 0.$$

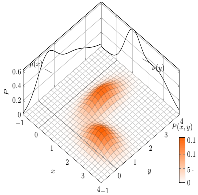


Image provided by Marco Cuturi

- In probabilistic terminology, a transport plan π is a **joint distribution** between μ and ν.

$\Pi(\mu, \nu)$ = the set of transport plans between μ and ν:

- **Transportation cost** $c(x, y)$ between x and y.
- Total transportation cost of a transport plan π:

$$\int_{X \times Y} c(x, y) d\pi(x, y).$$

- **The optimization problem:** Find the optimal solution $\bar{\pi}$ for

$$\inf_{\pi \in \Pi(\mu,\nu)} \int_{X \times Y} c(x, y) d\pi(x, y).$$

Brenier's theorem: characterization of optimal transport

- One of the most important theorems in optimal transport. It roughly says for the transportation cost function $c(x, y) = |x - y|^2/2$,

  "An optimal transport plan is given by a mapping $T = \nabla\phi$ for a convex function ϕ."

- For **more general type of cost** c, there is a generalization of this, called **Gangbo-McCann theorem**.

  "Under appropriate conditions on c, the optimal transport plan is given by a mapping T, which is the c**-gradient** of a c**-convex** function."

  - For example, $c(x, y) = dist(x, y)^p$, $p > 1$, $dist(x, y) =$ Euclidean/Riemannian distance, etc. ...

Optimal transport is a simple concept and thus has connections to many things!

A few research directions (among many) where ideas of optimal transport are applied:

- **Connecting different distributions in an effective way**
  - Developmental curves
- **Matchings under stochastic constraints**
  - Matching probability measures in stochastic ways
- **Providing solutions to partial differential equations**
  - Dynamics of phase transition/ free boundaries

Connecting different distributions in an effective way:

Optimal transport can be used to connect (many) different distributions in an effective way, and can be applied to the problem of **inferring**

- developmental curves

from sample data.

Evolution of distributions, in the gene expression space

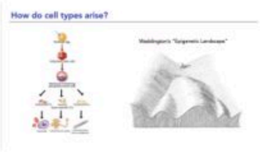


Figure courtesy of Geoffrey Schiebinger

One can view the developmental process of cells as the **evolution of distributions**, $\mathbf{P}_t$, $0 \leq t \leq t_{max}$, on the gene expression space.

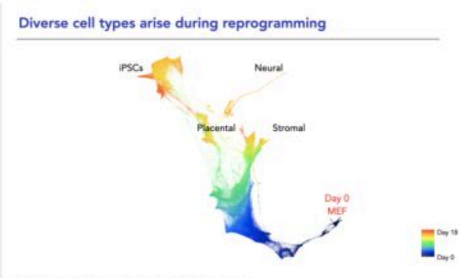
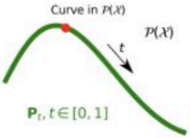


Figure courtesy of Geoffrey Schiebinger



The marginals $\mathbf{P}_t$ describe a curve in $P(\mathcal{X})$, the space of probability measures on $\mathcal{X}$.

How do we **see** the evolution trajectories of the cells?

- **Fundamental Difficulty:** To measure the gene expression, the cell is destroyed.
- Therefore, we cannot actually see the development of a cell in terms of the evolution of its gene data.
- So, the developmental process of a cell cannot be measured directly.

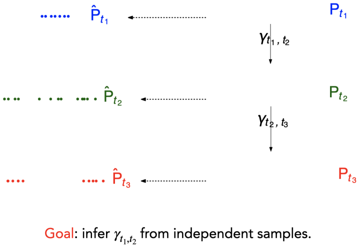
- We only see **snapshots of populations in time**, from the **samples** of different but identical populations.
- The individual trajectories for cells need to be **inferred** from those snapshots.

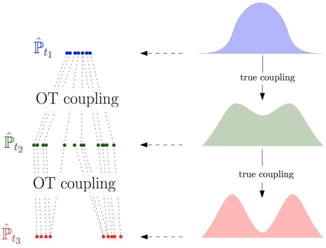
- This is a statistical inference problem:
  - Given a collection of samples from a distribution, can we guess what the distribution looks like?
  - Th more samples, the better our guess?

- **Additional difficulties:**
  - So many cells. And it is costly to test many cells.
  - So many genes. The gene expression space has dimension $\geq 20,000$.

Waddington-OT

- Optimal transport provides a good method for such an inference problem.





Figures courtesy of Geoffrey Schiebinger

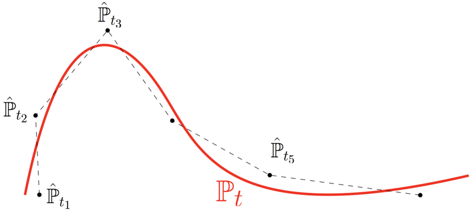
- Find optimal solutions $\hat{\pi}_i \in \Pi(\hat{\mathbf{P}}_{t_i}, \hat{\mathbf{P}}_{t_{i+1}})$.
  - It is about how each cell at time t_i develops to the next gene expression at time t_{i+1}.
- Then, glue together these couplings. $\hat{\pi}_T \circ \hat{\pi}_{T-1} \circ \cdots \circ \hat{\pi}_1 \circ \hat{\pi}_0$.

Waddington-OT

When we know only samples $\hat{\mathbf{P}}_{t_1}$ of $\mathbf{P}_{t_i}$, for samples of time $0 = t_0 < t_1 < ... < t_T = t_{max}$:

Intuition
In the space of distributions with the OT metric,
the process is locally linear:

Waddington-OT [Schiebinger et al. 2019]



It can approximate better and better.

- By increasing the number of time points T.
- By increasing number of samples for $\hat{\mathbf{P}}_{t_i}$.

Drawback: If the sample errors are big (e.g. due to small sample size), the solution may suffer overfitting, making it far from the ground truth.

- An extreme such case is when each sample (at a time point) is a single cell, even if there are many time points.

Optimal transport assumption

Waddington-OT is based on optimal transport assumption of the development.

- One can directly consider finding a law **P**,
  a probability measure on the space of paths,
  that describes the development of each cells over time.
  - It is a transport plan in continuous time:
    the particles (i.e. cells) are transported following a certain set of paths over time.

Assumption:
The development follows a law **P**, a probability measure on the space of paths, corresponding to an SDE of the form

$$dX_t = \nabla\Psi(t, X_t)dt + \sigma dB_t$$

- This assumption is a short time (dt) and stochastic version (dB_t) of the famous Brenier's theorem, which roughly says

  "An optimal transport plan is given by a mapping $T = \nabla\phi$ for a convex function ϕ."

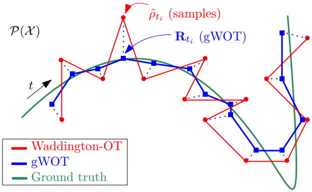
Question: How do we recover **P** from knowing the samples of marginals $\hat{\mathbf{P}}_{t_1}, \ldots, \hat{\mathbf{P}}_{t_T}$, especially when each sample size of $\hat{\mathbf{P}}_{t_i}$ is small so subject to significant error?
How to avoid overfitting?

Global Waddington OT (gWOT)

[Lavenant, Zhang, K., & Schiebinger, to appear 2023.]

Let the optimization also find the marginals approximating $\mathbf{P}_{t_i}$'s from the data $\hat{P}_{t_i}$'s.

- Combine '**regularization**' and '**data fitting**' together in the functional.
- $\mathbf{R}$ = a probability measure on the space of paths
  "=" a transport plan in continuous time.
- $\mathbf{R}_{t_i}$ = the marginal distribution of $\mathbf{R}$ at time t_i "=" the distribution of particles at time t_i that follow the paths governed by the law $\mathbf{R}$.
- $$\min_{\mathbf{R}} \left[\underbrace{\mathrm{Reg}(\mathbf{R})}_{\text{penalize overfitting}} + \frac{1}{\lambda} \underbrace{\mathrm{DataFitting}(\mathbf{R}_{t_1}, \ldots, \mathbf{R}_{t_T})}_{\text{penalize deviating from the data}} \right]$$
- λ = the balance between 'regularization' and 'data fitting'; it is to be guessed.
  - One may try different values of λ to find the 'best' result.



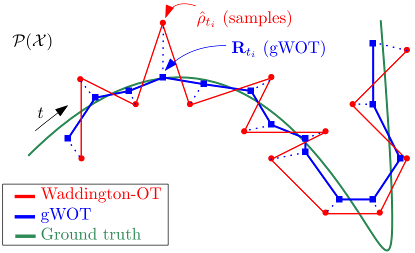
Global Waddington OT (gWOT): consistency

Theorem (Lavenant, Zhang, K., & Schiebinger)

Let $\mathbf{R}^{T,\lambda,h}$ denote the optimal solution of

$$\min_{\mathbf{R}} \left[\underbrace{\mathrm{Reg}(\mathbf{R})}_{\textit{penalize overfitting}} + \frac{1}{\lambda} \underbrace{\mathrm{DataFitting}(\mathbf{R}_{t_1}, \ldots, \mathbf{R}_{t_T})}_{\textit{penalize deviating from the data}} \right].$$

Then,
in the limit $T \to \infty$, followed by $\lambda \to 0$, $h \to 0$,
we have that $\mathbf{R}^{T,\lambda,h}$ converges to $\mathbf{P}$ (narrowly).



- It works even when each sample (at a time point) is a single cell, if there are many (thousands) time points.

Additional information: we skip this

- The relative entropy: $H(\alpha|\beta) = \int_\Omega \log\left(d\alpha/d\beta\right) d\alpha$.
  - $H(\alpha|\beta)$ measures how much α deviates from β.
- $$Reg(\mathbf{R}) = \sigma^2 H(\mathbf{R}|\mathbf{W}^\sigma)$$

  Compare the process $\mathbf{R}$ (a probability measure on the space of paths) with $\mathbf{W}^\sigma$ (the Wiener measure corresponding to the Brownian motion σB_t with diffusivity σ^2.)
- $$\text{DataFitting}(\mathbf{R}_{t_1}, ..., \mathbf{R}_{t_T}) = \sum_{i=1}^{T} |t_{i+1} - t_i| \left[H(\hat{\rho}^h_{t_i}|\mathbf{R}_{t_i}) - H(\hat{\rho}^h_{t_i}|\,\mathrm{vol})\right]$$

  $\hat{\rho}^h_{t_i}$ = a smooth approximation of the empirical data $\hat{P}_{t_i}$ at time t_i.

Further questions:

- Rate of convergence as we have more and more samples?
- What is the correct metric (equivalently, cost c), for the space of gene expressions?
  - In the previous work we used $c(x, y) = |x - y|^2/2$ for $x, y \in \mathbb{R}^d$ for the underlying distance structure on the gene expression space.

Matching under stochastic constraints

- Optimal transport reveals certain extremal structures of the matchings.

Matching under stochastic constraints

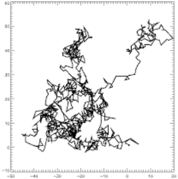
We may consider matching μ and ν where mass particles are moved in a certain restricted way:

- Constrain the transport to be a **martingale**:
  - **Question:** Given two probability measures μ and ν, can we realize ν by a martingale starting from μ, and if so what is the structure? Especially, if we do it in an optimal way?

- Constrain them to follow **stochastic processes**, e.g. **the Brownian motion**.
  - **Question:** Given two probability measures μ and ν,

    - can we realize ν by stopping appropriately the Brownian motion from the distribution μ,

    - and if so what are the properties of such a stopping time?
    Especially, if we do it in an optimal way?



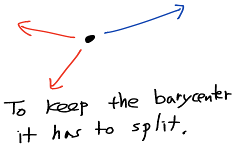
from CRM-physmath

Martingale

Branches out while keeping the barycentre.
(martingale constraint):

$$\int y\, d\pi_x(y) = x.$$

π is martingale $\Longrightarrow$ for π to move a unit mass, it has to **split the mass**!

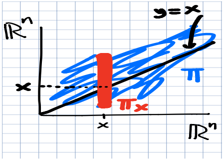


Remark: [Strassen]

"There is a martingale transport from μ to ν."

$\Leftrightarrow$ μ and ν are in convex order;
$\mu \leq_c \nu$, i.e. $\int \xi d\mu \leq \int \xi d\nu$, $\forall$ convex $\xi : \mathbb{R}^n \to \mathbb{R}$.

Optimal **Martingale** Transport Problem



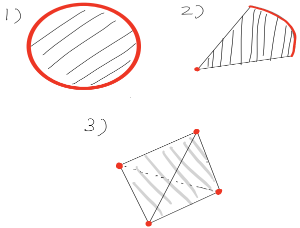
- $MT(\mu,\nu)$ = the set of margingale transport plans from μ to ν
-
$$\inf_{\pi \in MT(\mu,\nu)} \int_{\mathbb{R}^n \times \mathbb{R}^n} c(x,y)\,d\pi(x,y).$$
- For π to move a unit mass, it has to **split the mass**!

To keep the barycenter
it has to split.

Q: What is the **structure** of the set where the **optimal** martingale splits mass?

- [Hobson & Neuberger] [Begilebock & Juillet] In $\mathbb{R}^1$: Geometric structures of splitting sets, e.g. splitting into two points for $c(x,y) = |x-y|$.
- [Ghoussoub, K. & Lim] In $\mathbb{R}^n$: Geometric structures of splitting sets; **extremal structures**
  - e.g. when dim $= 2$
  - e.g. in general dimensions if ν can be obtained by a diffusion process from μ.

Further questions



Assume:

- In $\mathbb{R}^n$. $c(x, y) = |x - y|$
- $\mu << \mathcal{L}^n$
- $\pi \in MT(\mu, \nu)$ be optimal.

Conjecture [Ghoussoub, K. & Lim]
Then, for μ almost every x,
the splitting set Γ_x
has **extremal** property,
that is, $\overline{\Gamma}_x = \mathrm{Ext}\left(\mathrm{conv}(\overline{\Gamma}_x)\right)$.

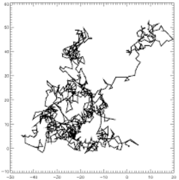
This conjecture is verified

- e.g. when $\mathrm{dim} = 2$;
- e.g. in general dimensions if ν can be obtained by a diffusion process from μ; or
- e.g. when the target measure ν is discrete.

Optimal transport of stopping Brownian motion

Question: Given two probability measures μ and ν,

- can we realize ν by stopping appropriately the Brownian motion from the distribution μ,
- and if so what are the properties of such a stopping time?
  Especially, if we do it in an optimal way?



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- A **stopping time** τ of Brownian motion is a random time at which one stops a particle following the Brownian motion; at that time, the decision to stop or not is made only using the information of the present and the past, not the future.
- Notice that stopping time depends on each particular path.
- For given nonnegative measures μ, ν and a stopping time τ,

$$\text{“}B_0 \sim \mu,\ B_\tau \sim \nu\text{”}$$

means that for each measurable set E,

$$\nu[E] = \int Prob[B_\tau \in E \mid B_0 = x] d\mu(x).$$

Skorokhod problem

For **given** probability measures μ, ν, does there exist a **stopping time** τ of the Brownian motion with $\mathbb{E}[\tau] < \infty$, such that

$$B_0 \sim \mu \quad \& \quad B_\tau \sim \nu?$$

- [Skorokhod] [Root] [Rost] [Azéma/Yor] [Vallois] [Perkins] [Jacka] ...[Obloj]...

Remark:

- For such a stopping time τ to exist
  we need
  - μ and ν are in **subharmonic** order, $\mu \prec_{SH} \nu$,
    i.e. $\int \xi d\mu \leq \int \xi d\nu$,
    $\forall$ subharmonic $\xi : \mathbb{R}^n \to \mathbb{R}$ $(\Delta\xi \geq 0)$.
- We may view the Brownian motion with a stopping time τ, from μ to ν, as a transport plan.

Connection to optimal transport: **Optimal Skorokhod Problem**

What can we say about an **optimal** stopping time τ for

$$\mathcal{P}(\mu,\nu) := \inf_{\tau}\{\, \mathcal{C}(\tau) \quad | \quad B_0 \sim \mu \quad \& \quad B_\tau \sim \nu\}?$$

- (Markovian cost) e.g. $\mathcal{C}(\tau) = \mathbb{E}\left[\int_0^\tau L(t, B_t)dt\right]$
- (non-Markovian cost) e.g. $\mathcal{C}(\tau) = \mathbb{E}\left[|B_0 - B_\tau|\right]$,

Literature:

- [Hobson] .. ....
- [Beiglböck/Cox/Huesmann '13].
  - **Optimal transport** unifies the previous results on Skorokhod problem, with Markovian cost.
- [Ghoussoub, K. & Palmer '18–'20]. PDE and dynamic programming methods: dual attainment, general dimensions, Markovian and non-Markovian cost, etc.
- And many more people.

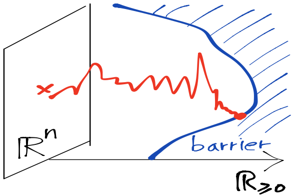
Optimal Skorokhod problem $\mathcal{P}(\mu, \nu)$

[Beiglböck, Cox, & Huesmann '13]
[Ghoussoub, K. & Palmer '18].

- geometric structures for the cost $\mathbb{E}\left[\int_0^\tau L(t, B_t)dt\right]$.
- The optimal stopping time is uniquely determined by hitting a certain **barrier in the space-time $\mathbb{R}_{\geq 0} \times \mathbb{R}^n$.**

- **Barrier** $R \subset \mathbb{R}_{\geq 0} \times \mathbb{R}^n$
- The **hitting time** τ^R to R,

$$\tau^R := \inf\{t \geq 0 \mid (t, B_t) \in R\}.$$



[Ghoussoub, K. & Palmer '18] determined the optimal barrier R^* by solving the optimal solution to the dual problem (on a certain space of functions).

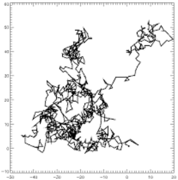
Further questions

- How can we analyze the properties of the optimal barriers R^*?
- The optimal barriers play the role of Brenier's optimal mapping for the stopping time problem.
  - For Brenier's optimal mapping, we have a corresponding PDE, called the Monge-Ampère equation to study its properties.
  - For the optimal barrier, what analytical tools can we develop?

Optimal transport approach to free boundary problems

Application of optimal transport methods to free boundary problems

- An optimal transport approach gives a robust method for studying free boundaries.
- Optimal transport with the Brownian motion stopping gives the description of the **free boundary** of phase transition phenomena. [Kim & K.]



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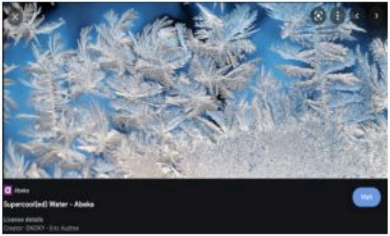
The Stefan Problem

Supercooled water freezing into Ice



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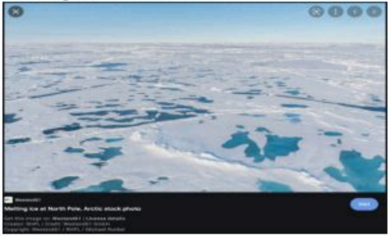
unstable behaviour

Melting of ice into water



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stable behavior

- Only few results on the supercooled problem **(the unstable problem)**:
  - 1-D, special case [Chayes/Swindle][Chayes/I.Kim'08][Lacoin '14][Delarue-Nadtochiy-Shkolnikov '19].
  - self-similar, well-prepared (smooth) radial data [Hadzic/Raphael]
- Many works for the **stable problem**
  - Well-posedness: [Meirmanov] .....
  - Regularity: [Caffarelli'77], [Athanasopoulos/Cafferlli/Salsa '96-98][Choi/I.Kim'10], ... recently, [Figalli, Cabre, Serra, Ros-Oton]...
  - Stability: [Hadzic/Shkoller'14][Hadzic/Raphael'16], ...
- The supercooled problem (the unstable problem) is poorly understood, while there are many good results for the stable problem.
- **The supercooled problem (the unstable problem) is not well-posed in general.** There is in general no uniqueness for the solutions, and little is known for its global existence especially in multi-dimensions.

Free target problem (K. & I. Kim)

- **[Free target under a density constraint]**
  Given $f \geq 0$, $f \in L^\infty(\mathbb{R}^n)$ (e.g. $f \equiv 1$).
  consider

$$\mathcal{P}_f(\mu) := \inf_{\tau,\nu}\{\mathcal{C}(\tau) \mid B_0 \sim \mu,\ B_\tau \sim \nu,\ \nu \leq f,\ \nu \text{ is compactly supported}\}$$

  Here $\mathcal{C}(\tau) = \mathbb{E}\left[\int_0^\tau L(t, B_t)dt\right]$, the transportation cost along the Brownian motion up to the stopping time τ.

- The optimal solution ν^* **saturates** the upper bound constraint f on the active region, namely,

$$\nu^*(x) = f(x) \text{ "if an active Brownian particle pass by at } x\text{".}$$

- The pair of optimal stopping and the optimal target (τ^*, ν^*) has corresponding moving mass η and stopping rate ρ in the space-time, **solving**

  the Stefan problem

$$\partial_t \eta - \frac{1}{2}\Delta\eta = -\rho$$
$$\rho = \pm\partial_t\left(f\,\chi_{\{\eta=0\}}\right) \leftarrow \text{This is due to the saturation result.}$$

  '$+$' for the supercooled (unstable) problem, '$-$' for the usual (stable) problem.
  - For the original problem $f \equiv 1$.
  - The boundary $\partial(\chi_{\{\eta=0\}})$ is the free boundary at a given time.

- Our method works for both stable and unstable (supercooled) problems.

- In particular, we are able to obtain a certain type of well-posedness result for **the supercooled Stefan in general dimensions**.
  - **Our result is a limited result:**
  - It solves the supercoolded (unstable) problem by **providing a good initial active domain** $E = \lim_{t \to 0^+} \{x \mid \eta(x, t) > 0\}$ determined by the optimal free target ν^*.

Further questions:

- **Our method is robust and can be applied to**

  Nonlocal Stefan problem:
  This is modelled on stochastic processes that allow jump of particles.
  As a PDE, it can be written:
  (for $0 < s < 1$),

$$\partial_t \eta + (-\Delta)^s \eta = -\rho$$

  where

$$\rho := \begin{cases} \partial_t \left((1 - \kappa_1) \chi_{\{\eta = 0\}} \right) & \text{(supercooled)}, \\ \partial_t \left((-1 + \kappa_2) \chi_{\{\eta = 0\}} \right) & \text{(the usual)}. \end{cases}$$

  - Here, κ_1, κ_2 arise due to nonlocal effect: the stochastic particles that carries heat can jump across the free boundary and stop.
  - Recent result [Chu, K., Kim, & Nam].
- **A difficult open problem** is to solve the supercooled Stefan problem **where the initial active domain** $E = \lim_{t \to 0^+} \{x \mid \eta(x,t) > 0\}$ **is prescribed**.
  - A result to be announced soon. [Choi, K. & Kim].
- Regularity of free boundary for a certain class of solutions?
  - This is related to developing methods for analyzing the optimal barriers in the optimal stopping time problems.

Kantorovich Initiative

https://kantorovich.org

Supported by **Pacific Institute for the Mathematical Sciences (PIMS).**

- This is an interdisciplinary initiative started by people in Pacific Northwest region in North America (Vancouver, Seattle, Edmonton, Saskatoon), and in various areas (mathematics, statistics, economics, computer science, biology) **connected by optimal transport theory**.
- **KI online seminar series** (sign up for mailing list!)
- Postdoc opportunities
- Conferences

Thank you!